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Thermal-Induced Microplastic Contamination from Food Packaging: Implications for Consumer Health in Northern Nigeria

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Abstract

Background: Microplastic ingestion through food packaging represents an emerging public health concern with particular relevance to developing economies where plastic usage patterns may increase exposure risks.

Objective: To quantify and characterize microplastic leaching from polypropylene and polystyrene food containers under simulated household heating conditions in Sokoto, Nigeria.

Methods: A controlled experimental study was conducted using commercially available PP and PS containers. Thermal exposure protocols mimicked common practices: boiling water immersion (PS) and oven heating (PP). Leachate analysis employed FTIR spectroscopy with blank-controlled methodology.

Results: Both polymer types released detectable microplastic residues. FTIR identification confirmed: PS—aromatic C-H (3025 cm^{-1}), C=C (1600 cm^{-1}), C-H bending (742 cm^{-1}); PP—aliphatic C-H (2950 cm^{-1}), CH_2 (1455 cm^{-1}), C-C (1167 cm^{-1}). PS demonstrated 2–3× greater spectral intensity, indicating higher leaching propensity.

Conclusion: Routine heating of food in plastic containers contributes to microplastic dietary exposure. Health promotion strategies should prioritize alternative container materials and safe usage guidelines.

Nigeria's food sector relies extensively on plastic containers for storage, transportation, and vending

Keywords: *Food safety, microplastics, thermal degradation, consumer exposure, Nigeria*

1. Background

The global proliferation of plastic food packaging has generated unprecedented environmental and health challenges. Microplastics particles <5mm have permeated ecosystems worldwide, with documented presence in human tissues including blood, placenta, and gastrointestinal tract (Prata et al., 2020). Food packaging constitutes a primary exposure vector, particularly when thermal or mechanical stress accelerates polymer degradation.

Nigeria's food sector relies extensively on plastic containers for storage, transportation, and vending. The combination of high ambient temperatures, repeated container reuse, and limited consumer awareness creates conditions conducive to elevated microplastic ingestion. This study provides foundational data to inform public health interventions.

2. Methodology

Study Design: Experimental simulation of household food storage practices

Materials: Commercial PP (microwaveable containers) and PS (disposable containers) obtained from Sokoto markets

Procedures:

a. PS protocol: 100°C distilled water, 3-hour contact, 3-hour standing
b. PP protocol: 80°C oven, 30-minute exposure
c. Blank controls: Identical conditions without containers

Analysis: **FTIR spectroscopy (4000–400 cm^{-1}) with polymer standard comparison**

3. Findings

Polymer	Key FTIR Peaks (cm^{-1})	Assignment	Relative Intensity
PS	3025	Aromatic C-H stretch	High
	1600	Aromatic C=C stretch	Moderate
	742	Aromatic C-H bend	High
PP	2950	Aliphatic C-H stretch	Moderate
	1455	CH_2 bending	Moderate
	1167	C-C stretching	Low-Moderate

4. Health Implications

Microplastic ingestion has been associated with:

a. Physical tissue damage and inflammation
b. Additive chemical transfer (plasticizers, stabilizers)
c. Gut microbiome disruption
d. Potential translocation across biological barriers

The higher leaching observed from PS containers, commonly used for hot beverages and street foods in Nigerian markets suggests disproportionate exposure risks for economically disadvantaged populations dependent on informal food vendors.

5. Recommendations

Policy Level:

a. Establishment of Nigerian Standards for food-contact plastics
b. Mandatory labeling of temperature limitations
c. Incentives for biodegradable alternatives

Public Health:

a. Community education on safe container usage
b. Promotion of traditional materials (clay, stainless steel)
c. Integration into school nutrition programs

Research Priorities:

a. Biomonitoring studies in Nigerian populations
b. Long-term health outcome tracking
c. Intervention effectiveness evaluation

10. Transparency Statement

No conflicts of interest were identified that could compromise scholarly objectivity.

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